Econometrics Midterm

R functions and output interpretation · 5 code exhibits · W2, W5, W6 · 23 questions

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Exhibit A · Lab2.R — descriptive statistics (W2)

Used in questions 1–4

```
1  practice <- read.csv("Lab2_Practice.csv", header=TRUE, na.strings=".")
2  competence <- practice$COMPETENCE
3  breaks <- practice$BREAKS
4  grade <- practice$GRADE
5  hist(competence, breaks=15)
6  summary(breaks)
7  sum((competence-mean(competence))^2)/(length(competence)-1)
8  var(competence)
9  quantile(breaks)
10 IQR(breaks)
11 boxplot(grade)
12 abline(h=median(grade), col="red")
13 par(mfrow = c(1, 2))
14 boxplot(grade)
15 vioplot(grade)
```

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1. [Exhibit A] Line 9. What does quantile() do?

2. [Exhibit A] Line 13. What does par(mfrow = c(1, 2)) do?

3. [Exhibit A] Lines 7–8. Both lines print 4.772. Why do they agree, and why does the formula divide by n − 1 and not by n?

4. [Exhibit A] Lines 9–10 (with line 6). The output for breaks is below. What do the mean and the median say about the shape of the distribution? Is the maximum an outlier by the 1.5 × IQR rule? Write the fence.

summary(breaks) and IQR(breaks) · n = 27

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	IQR
0	1	2	3.185	4.5	12	3.5

Exhibit B · Lab4.R — functions and strings (W2)

Used in questions 5–6

```
1  library(readxl)
2  library(stringr)
3  to12 <- function(str){
4    hour=as.integer(str)
5    if (hour>12) {
6      return(paste(hour-12, "PM"))
7    } else {
8      return(paste(hour, "AM"))
9    }
10 }
11 hasMultipleLoc <- function(str){
12   str_detect(str, fixed(","))
13 }
14 data=read_excel("Café.xlsx")
15 data <- as.data.frame(data)
16 colnames(data)=c("type","location","openingHour","closingHour")
17 (openingHour <- sapply(data$openingHour, to12))
18 data$openingHour=openingHour
19 (locationText <- ifelse(sapply(data$location, hasMultipleLoc), "Yes", "No"))
20 data$hasMultipleLoc=locationText
```

5. [Exhibit B] Line 12. What does str_detect() do, and what is fixed() for?

6. [Exhibit B] Line 17. What does sapply() do?

Exhibit C · regression.R (condensed) — OLS, inference and prediction (W5)

Used in questions 7–12

```
1  social <- read.csv("social.csv", header=T)
2  perform=social$cryp_scoreFINAL
3  eye=social$RMET_mean
4  skill=social$Smean_mean
5  ravens=social$ravens_mean
6  plot(perform ~ eye)
7  eye_regr=lm(perform ~ eye)
8  abline(eye_regr, col="blue")
9  abline(h=mean(perform), col="red")
10 abline(v=mean(eye), col="green")
11 skill_regr=lm(perform ~ skill)
12 skill_summary=summary(skill_regr)
13 confint(skill_regr, level=0.99)
14 (skill_coef=cor(perform,skill)^2)
15 round(skill_summary$r.squared-skill_coef,16)
16 mult_lm=lm(perform ~ skill + ravens)
17 (mult_summary=summary(mult_lm))
18 predict(mult_lm, data.frame(skill=c(0.5,1.5), ravens=c(1.5,0.5)), interval="confidence",
   level=0.95)
19 points <- seq(min(eye)-3, max(eye)+3, length.out=10000)
20 pred <- predict(eye_regr, data.frame(eye=points), interval="confidence", level=0.95)
21 polygon(c(rev(points), points), c(rev(pred[ ,3]), pred[ ,2]), col=rgb(0,0,0,0.1),
   border=NA)
```

7. [Exhibit C] Line 13. What does confint() do, and what does level = 0.99 change?

8. [Exhibit C] Line 18. What does predict() do, and what does interval = "confidence" add?

9. [Exhibit C] Lines 11–13. Read the output of `summary(skill_regr)` and `confint()` below (all variables are standardized). Interpret the slope, say whether it is significant at 5% and at 1% and how the two intervals show this, and say what share of the variation in perform skill explains.

summary(skill_regr) and confint() · n = 85

Term	Estimate	Std. Error	t value	p-value	95% CI	99% CI
(Intercept)	0.0000	0.1066	0.000	1.0000	[−0.212, 0.212]	[−0.281, 0.281]
skill	0.2141	0.1072	1.997	0.0491	[0.0009, 0.4274]	[−0.0685, 0.4968]
Multiple R-squared 0.04585 · Adjusted R-squared 0.03436 · F 3.989 on 1 and 83 DF, p = 0.0491						

10. [Exhibit C] Lines 16–17. Read the output of `summary(mult_lm)` below. Interpret the skill coefficient. Then explain how neither slope is significant at 5% on its own while the regression as a whole is.

summary(mult_lm) · n = 85

Term	Estimate	Std. Error	t value	p-value
(Intercept)	0.0000	0.1056	0.000	1.000
skill	0.1712	0.1096	1.562	0.122
ravens	0.1746	0.1096	1.593	0.115
Multiple R-squared 0.07449 · Adjusted R-squared 0.05192 · F 3.30 on 2 and 82 DF, p = 0.04184				

11. [Exhibit C] Lines 11 and 16. The skill slope is 0.2141 in the simple regression (line 11) and 0.1712 once ravens is added (line 16). The slope from regressing ravens on skill is 0.2458. Show how these numbers fit together and explain the drop.

12. [Exhibit C] Line 18. The output below also shows `interval = "prediction"` for the same two new teams. Interpret the first confidence interval, and explain why the prediction interval is so much wider.

predict() · 95%

New team	fit	conf. lwr	conf. upr	pred. lwr	pred. upr
skill 0.5, ravens 1.5	0.3475	−0.0339	0.7289	−1.6267	2.3217
skill 1.5, ravens 0.5	0.3441	−0.0373	0.7255	−1.6301	2.3183

Exhibit D · simple regression demonstration.R (condensed) — simulation of tests and intervals (W5)

Used in questions 13–17

```
1  set.seed(4201)
2  slope=0.2; intercept=3; sampleSize=10; sampleCnt=10000
3  estimator=c(); t=c(); pValue=c()
4  for(rowIdx in 1:sampleCnt) {
5    fertilizer=rnorm(sampleSize,10,2)
6    yield=intercept+slope*fertilizer+rnorm(sampleSize,0,1)
7    coef=summary(lm(yield~fertilizer))$coefficients
8    estimator[rowIdx]=coef[2,1]
9    t[rowIdx]=coef[2,3]
10   pValue[rowIdx]=coef[2,4]
11 }
12 hist(estimator)
13 alpha=0.05
14 rejection=ifelse(pValue<=alpha, "Reject", "Fail to Reject")
15 (result=table(rejection))
16 barplot(result, main="Null Hypothesis Rejection")
17 sampleSize=12; slope=2; sampleCnt=20
18 results=data.frame(index=1:sampleCnt)
19 for(rowIdx in 1:sampleCnt) {
20   fertilizer=rnorm(sampleSize,10,2)
21   yield=intercept+slope*fertilizer+rnorm(sampleSize,0,1)
22   fer_regr=lm(yield ~ fertilizer)
23   results[rowIdx,"slope"]=fer_regr$coefficients[2]
24   interval=confint(fer_regr)
25   lowerBound=interval["fertilizer","2.5 %"]
26   upperBound=interval["fertilizer","97.5 %"]
27   results[rowIdx,"inside"]=lowerBound<=slope && slope<=upperBound
28 }
```

13. [Exhibit D] Line 1. What does set.seed() do?

14. [Exhibit D] Line 14. What does ifelse() do?

15. [Exhibit D] Lines 4–12. Over the 10,000 samples, mean(estimator) = 0.2027 and sd(estimator) = 0.190, while the true slope is 0.2. What property of OLS does this show, and what does the histogram on line 12 picture?

16. [Exhibit D] Line 15. The table prints Fail to Reject 8092 and Reject 1908, and the true slope is 0.2, not 0. What does the 19.1% rejection rate measure, what is the other 80.9%, and what would the rate be if the true slope were 0?

table(rejection) · 10,000 samples

Fail to Reject	Reject
8092	1908

17. [Exhibit D] Lines 17–28. In this run all 20 intervals contain the true slope 2 (inside = TRUE 20 times). How many would you expect, and what does the 95% level actually promise?

Exhibit E · W6 — dummy variables and interactions (lecture examples, Wooldridge Ch.7)

Used in questions 18–23

```
1 m1 <- lm(wage ~ female + educ, data = df)
2 m2 <- lm(wage ~ educ*female, data = df)
3 m3 <- lm(wage ~ educ + as.factor(region), data = df)
4 library(car)
5 linearHypothesis(m2, c("female = 0", "educ:female = 0"))
6 restricted <- lm(wage ~ educ + exper, data = df)
7 unrestricted <- lm(wage ~ (educ + exper)*female, data = df)
8 anova(restricted, unrestricted)
9 lpm <- lm(inlf ~ educ + age, data = df)
```

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18. [Exhibit E] Line 3. What does as.factor() do, and what does lm() then create from it?

19. [Exhibit E] Line 8. What does anova() do when it is given two models?

20. [Exhibit E] Line 1, estimated with df = wage1 (Wooldridge, n = 526), gives predicted wage = 0.623 − 2.273 female + 0.506 educ, with a standard error of 0.279 on female. Interpret the female coefficient. Which group is the base, and why can't a male dummy be added as well?

21. [Exhibit E] Lines 2 and 5. The output of m2 (wage1) is below. What is the estimated return to education for men and for women, and is the gap significant? Line 5 tests female = 0 and educ:female = 0 together and gives F = 33.51, p < 0.001. Why can it reject when neither t-test does?

summary(m2) · wage1, n = 526

Term	Estimate	Std. Error	t value	p-value
(Intercept)	0.2005	0.8436	0.238	0.812
educ	0.5395	0.0642	8.400	<0.001
female	−1.1985	1.3250	−0.905	0.366
educ:female	−0.0860	0.1036	−0.830	0.407

22. [Exhibit E] Line 8. The anova() table (wage1) is below. Compute F from the RSS values and state the conclusion. What does the test assume about the error variance?

anova(restricted, unrestricted) · wage1

Model	Res.Df	RSS	Df	Sum of Sq	F	p-value
restricted	523	5548.2				
unrestricted	520	4737.8	3	810.38	29.648	<2.2e-16

23. [Exhibit E] Line 9 is a linear probability model (inlf = 1 if a woman is in the labor force). The textbook version (Example 7.12, mroz, n = 753) is below. Interpret the educ and kidslt6 coefficients, and name two weaknesses of the LPM that this fit shows.

```
lpm <- lm(inlf ~ nwifeinc + educ + exper + I(exper^2) + age + kidslt6 + kidsge6,
data = mroz)
```

Selected coefficients · fitted values from −0.345 to 1.127

Term	Estimate	Std. Error	t value
educ	0.0380	0.0074	5.15
kidslt6	−0.2618	0.0335	−7.81
age	−0.0161	0.0025	−6.48
nwifeinc	−0.0034	0.0014	−2.35

Econometrics Midterm — Model answers

1. W2 · quantile()
quantile(): returns sample quantiles, the values below which given shares of the data fall
By default it gives five numbers, the 0%, 25%, 50%, 75% and 100% points: the **minimum**, the first **quartile**, the median, the third quartile and the **maximum**
Other points can be asked for with probs, for example quantile(breaks, probs = 0.9)
Key results: quartile · 0% · minimum

2. W2 · par(mfrow =)
par(): sets **graphics parameters** for the plots that follow
mfrow = c(1, 2) splits the plotting window into **1 row and 2 columns**, filled row by row
So the next two plots (lines 14–15) appear **side by side**, until the layout is reset with par(mfrow = c(1, 1))
Key results: graphics parameter · 1 row and 2 columns · side by side

3. W2 · Ch.2 sample variance
Line 7: computes the sample variance by hand, $s^2 = \Sigma(x_i - \bar{x})^2 \div (n - 1) = 4.772$ for competence (n = 27)
var(): uses exactly the same formula, so both lines print **4.772**
Why n − 1: $\bar{x}$ is estimated from the same data and uses up one **degree of freedom**, and dividing by n − 1 makes s^2 an **unbiased** estimator of σ^2
Same idea in Ch.3: $\sigma^2 = SSR \div (n - k - 1)$
Key results: 4.772 · var() · n - 1 · unbiased

4. W2 · Ch.2 output: summary() and IQR()
Mean vs median: mean 3.185 > median 2, so a few large values pull the mean up and the distribution is **right-skewed**
IQR = Q3 − Q1 = 4.5 − 1 = 3.5
Upper fence = Q3 + 1.5 × IQR = 4.5 + 1.5 × 3.5 = **9.75**
Max 12 > 9.75, so 12 is an **outlier** and boxplot() draws it as a separate point
Robustness: the median and the IQR are not moved by it, the mean and the variance are
Key results: right-skewed · 9.75 · outlier

5. W2 · str_detect()
str_detect(): checks whether each string **contains** a pattern and returns **TRUE** or FALSE for each one
fixed(): makes the pattern a **literal** string, not a regular expression
So hasMultipleLoc("Seoul, Suwon") is TRUE and hasMultipleLoc("Seoul") is FALSE
Key results: TRUE · contains · literal

6. W2 · sapply()
sapply(): **applies** a function to **each element** of a vector or list
It then simplifies the results into a **vector** (or a matrix) when it can
Here it runs to12() on every opening hour, which to12() cannot do by itself because its if() takes one value at a time
Key results: each element · applies · vector

7. W5 · confint()
confint(): returns a **confidence interval** for each **coefficient** of a fitted model, a lower and an upper bound per term
By default the level is 95%
level = 0.99 gives 99% intervals, which are **wider**
Key results: confidence interval · coefficient · wider

8. W5 · predict()
predict(): plugs **new data**, a data frame with the model's variable names, into the fitted equation and returns the **predicted** values, here about 0.3475 and 0.3441
interval = "confidence" adds the **lower and upper** bounds, so the result has the columns fit, lwr and upr
Key results: predicted · new data · lower

9. W5 · Ch.4 output: t-test, CI and R^2
Slope: one standard deviation more social skill goes with **0.214** SD higher team performance
t = 0.2141 ÷ 0.1072 = **1.997** with df = 85 − 2 = 83
p = **0.049** < 0.05, so H_0: $\beta_1 = 0$ is rejected at 5% but not at 1% (p > 0.01)
Intervals: the 95% CI [0.0009, 0.4274] excludes 0 and the **99%** CI [−0.0685, 0.4968] includes 0, because a two-sided test at level α rejects exactly when the 100(1 − α)% CI excludes the null value
R^2 = 0.04585: skill explains about **4.6%** of the variation, and in simple regression R^2 = cor(perform, skill)^2, which is why lines 14–15 print a difference of 0
Low R^2: does not make the slope useless, it can still be a precise ceteris paribus estimate (Ch.3)
Key results: 0.214 · 1.997 · 0.0491 · 99% · 4.6%

10. W5 · Ch.4 output: separate t-tests vs the F-test
Skill: holding ravens fixed (**ceteris paribus**), one SD more social skill goes with 0.171 SD higher performance
t-tests: p = **0.122** (skill) and 0.115 (ravens), so neither is significant at 5% on its own
F-test: H_0: $\beta_1 = \beta_2 = 0$ against H_1: at least one ≠ 0 gives F(2, 82) = 3.30 with p = **0.042** < 0.05, so the two are **jointly** significant
Why both can hold: each t-test asks whether one variable adds anything once the other is already in, and skill and ravens share explanatory power (correlation 0.246), so each looks weak alone while together they clearly explain performance
Adjusted R^2 = 0.0519 > 0.0344 (skill alone): adding ravens is worth the lost degree of freedom, as expected because its |t| = 1.59 > 1 (Ch.6)
Key results: holding ravens fixed · 0.122 · 0.0418 · jointly

11. W5 · Ch.3 omitted variable bias
Formula: simple slope = $\hat{\beta}_1 + \hat{\beta}_2 \times \tilde{\delta}_1$ (Ch.3)
Numbers: 0.1712 + 0.1746 × 0.2458 = 0.1712 + 0.0429 = **0.2141**
Why: ravens raises performance ($\hat{\beta}_2$ = **0.1746** > 0) and is positively correlated with skill ($\tilde{\delta}_1$ = **0.2458** > 0), so without ravens the skill slope picks up part of the ravens effect
Result: the simple slope is biased **upward** by 0.043, the **omitted variable** bias
Note: $\tilde{\delta}_1$ equals the correlation here because both variables are standardized
Key results: 0.2141 · 0.1746 · 0.2458 · omitted variable · upward

12. W5 · Ch.6 output: confidence vs prediction interval
Fit: $\hat{y}$ = 0.3475 for skill = 0.5 and ravens = 1.5
Confidence interval: [−0.034, 0.729] is a 95% range for the **average** performance E(y | x) of all teams with these scores, and it reflects only the uncertainty in the estimated coefficients
Prediction interval: [−1.627, 2.322] is a 95% range for **one** new team, and it adds the variance of that team's own **error** u, se($\hat{y}$)^2 + $\hat{\sigma}^2$ with $\hat{\sigma}$ = 0.974
Why wider: $\hat{\sigma}^2$ dominates and does not shrink as n grows, so the prediction interval is always **wider** (Ch.6)
Key results: 0.3475 · average · individual · error · wider

13. W5 · set.seed()
set.seed(): sets the starting point (seed) of the **random number generator**
Every run with the same seed draws the **same random numbers**, so the simulation is **reproducible**
Without it, rnorm() would give different samples on each run
Key results: reproducible · random number generator · same random numbers

14. W5 · ifelse()
ifelse(): checks a **condition** for **each element** of a vector
Where it is TRUE it returns the **second** argument, and where it is FALSE the third
Here every p-value becomes "Reject" (p-value at most 0.05) or "Fail to Reject"
Key results: condition · each element · second

15. W5 · Ch.3 unbiasedness in the simulation
Mean: 0.2027 ≈ 0.2, so across repeated samples OLS is right on average, $E(\hat{\beta}_1) = \beta_1$: **unbiasedness** (MLR.1–MLR.4)
hist(estimator): pictures the **sampling distribution** of $\hat{\beta}_1$, centred at 0.2 and roughly normal because the errors are normal (MLR.6)
Spread: sd 0.190 is the sampling standard deviation, and it shrinks with a larger n or more variation in fertilizer, since $Var(\hat{\beta}_1) = \sigma^2 \div SST_x$
Key results: unbiased · 0.2027 · sampling distribution

16. W5 · Ch.4 output: rejection rate = power
Power: H_0 ($\beta_1 = 0$) is false here, so every rejection is correct and 1908 ÷ 10,000 = **19.1%** estimates the **power** of the 5% t-test
Type II: the other 80.9% fail to reject a false H_0, a **Type II** error
Why low: the effect is small relative to the noise and n is only 10, so a larger slope or a larger n raises power
True slope 0: H_0 is then true and the rejection rate is about α = **5%**, the Type I error rate (a rerun with slope = 0 gave 5.2%)
Key results: power · 19.1% · type II · 5%

17. W5 · Ch.4 output: what "95% confident" means
Expected: 0.95 × 20 = **19** intervals contain the true slope, and 20 of 20 is well within normal sampling variation
Meaning: 95% is a property of the method, in **repeated samples** 95% of the intervals built this way cover the **true slope**
Single interval: once computed it either contains 2 or it does not, so "a 95% probability that β_1 lies in this interval" is the wrong reading
Key point: the bounds are **random**, β_1 is fixed (Ch.4)
Key results: 19 · repeated samples · true slope · random

18. W6 · as.factor()
as.factor(): converts a variable into a **factor**, a **categorical** variable with levels, instead of a number
lm(): then creates one **dummy** for each level except the **base** (reference) level
Each dummy coefficient is the difference from the base level
Key results: categorical · dummy · base

19. W6 · anova()
anova(): **compares** two **nested** models with an **F-test**
It uses the residual sum of squares (**RSS**) of each model and the number of extra parameters, and reports F and its p-value
Here the models differ by female and its interactions, so it is the **Chow** test
Key results: nested · F-test · RSS · Chow

20. W6 · Ch.7 output: dummy coefficient
female: among people with the **same education**, women earn about **$2.27** per hour less than men on average (t = −2.273 ÷ 0.279 = −8.15, p < 0.001)
Base group: men (female = 0), the intercept 0.623 belongs to the men's line and −2.273 shifts it down for women
Male dummy: female + male = 1 for everyone, an exact linear relation with the intercept, the **dummy variable trap** (perfect collinearity, MLR.3)
Caution: this is not proof of discrimination, since female may be correlated with omitted productivity factors such as experience (Ch.7)
Key results: 2.27 · same education · base · dummy variable trap

21. W6 · Ch.7 output: interaction and joint test
Men: return to education = **0.539** dollars per hour for each year (female = 0)
Women: 0.5395 − 0.0860 = **0.4535**
Gap: educ:female = −0.086 with p = **0.407**, so no evidence that the return differs
Joint test: F = **33.51** with p < 0.001 rejects H_0 that men and women have the same wage line
Why: female and educ:female are almost collinear (correlation 0.96), so their separate t-tests are weak while **jointly** they clearly matter
Reading: women earn less, the data just cannot say whether it is the intercept or the slope (at the mean educ 12.56 the gap is −1.1985 − 0.086 × 12.56 ≈ −2.28)
Key results: 0.539 · 0.453 · 0.407 · 33.5 · jointly

22. W6 · Ch.7 output: Chow test by hand
Formula: $F = [(SSR_r - SSR_{ur}) \div q] \div [SSR_{ur} \div (n - k - 1)]$
Numbers: $[(5548.2 - 4737.8) \div \mathbf{3}] \div (4737.8 \div \mathbf{520}) = 270.1 \div 9.111 = \mathbf{29.65}$
Restrictions: q = 3, the female intercept shift and its two slope interactions, and df = 526 − 6 = 520
Conclusion: p < 2.2e-16, so **reject** H_0 that men and women share one wage equation
Assumption: the Chow test needs the **same error variance** in both groups (Ch.7)
Key results: 29.6 · 3 · 520 · reject · same variance

23. W6 · Ch.7 output: linear probability model
educ: one more year of education raises the probability of being in the labor force by 0.038, that is **3.8 percentage points**, ceteris paribus
kidslt6: each child under 6 lowers it by **26.2** percentage points
Weakness 1: fitted values run from −0.345 to 1.127, and 33 of 753 (16 below 0, 17 above 1) fall **outside** [0, 1]
Weakness 2: Var(u | x) = p(x)[1 − p(x)] depends on x, so the LPM is necessarily **heteroskedastic** and needs robust standard errors
Also: a constant effect cannot hold everywhere, since going from 0 to 4 young children would lower the probability by 4 × 0.262 = 1.05
Key results: percentage points · 3.8 · 26.2 · below 0 · heteroskedastic